Project Report: DevSync — Collaborative Developer Issue Tracker

1. Project Overview

DevSync is an internal developer productivity tool that allows employees to raise coding issues, share code snippets with error context, and collaborate with teammates in real time.

Most existing tools require developers to leave their coding environment to ask for help — via WhatsApp, email, or Slack. DevSync solves this by bringing collaboration directly into the developer's workflow, first as a web application and later as a VS Code Extension.

2. Why This Project Is Different

Existing Approach:

- Developers share code by copying it into WhatsApp or Slack manually
- No structured way to raise, assign, or track coding issues internally
- No history of past errors and how they were resolved
- No visibility into who is available to help at a given moment

Proposed System:

- Structured issue-raising system with code snippet and error message support
- Assign issues directly to a teammate from inside the app
- Real-time ping and code sharing via VS Code Extension (Phase 2)
- AI-powered error analysis and smart assignment suggestions (Phase 3)

This system **does not replace existing communication tools** — it complements them by adding a structured, developer-native layer for issue tracking and code collaboration.

3. Problem Statement

Currently, there is no internal system at Yash Technologies for developers to manage coding issues collaboratively. This results in:

- Time wasted explaining errors without proper context
- No tracking of who is working on which issue
- Repeated errors with no shared knowledge base
- Managers with no visibility into team-level coding blockers

4. Project Objectives

- Provide a structured platform for raising and tracking coding issues
- Allow developers to share code snippets and error messages with teammates
- Enable real-time collaboration through a VS Code Extension
- Give managers a dashboard to monitor team activity and issue resolution
- Integrate AI to automatically analyze errors and suggest fixes

5. System Workflow

1. **Issue Raised:** Developer encounters an error and raises an issue with title, description, code snippet, and error message
2. **Assignment:** The developer assigns the issue to an available teammate
3. **Collaboration:** Assigned teammate reviews the issue and provides a solution via the web app or VS Code Extension
4. **Resolution:** Issue is marked as Resolved and stored in history for future reference
5. **AI Assist (Phase 3):** AI analyzes the error and suggests a fix automatically

6. Technology Stack

- **Python:** Core backend logic and API development
- **Streamlit:** Web-based dashboard for the application
- **MySQL:** Permanent data storage for issues, users, and activity
- **FastAPI:** REST API backend connecting frontend and database
- **JavaScript + VS Code API:** VS Code Extension for real-time collaboration (Phase 2)
- **WebSocket:** Real-time messaging between developers (Phase 2)
- **Groq API (llama-3.3-70b):** AI-powered error analysis and suggestions (Phase 3)
- **scikit-learn:** Smart assignment model based on past issue data (Phase 3)

7. Future Scope

DevSync is designed to be scalable and can be enhanced further:

- **GitHub Integration:** Auto-link issues to commits and pull requests
- **Slack / Teams Bot:** Receive DevSync notifications in company messaging tools
- **Voice Issue Reporting:** Speak your error, AI logs the issue automatically
- **Mobile App:** React Native version for on-the-go issue tracking
- **Analytics Dashboard:** Team productivity reports, bottleneck detection, resolution time tracking
- **Multi-company SaaS:** Deployable as a product for other IT companies

8. Conclusion

DevSync addresses a real, daily operational problem faced by every software development team. By starting with a solid web application and progressively evolving it into a VS Code Extension with AI capabilities, the project demonstrates not just technical skill but also a deep understanding of developer workflows.

The system starts with a simple, functional implementation and offers strong potential for future enhancements using advanced technologies — making it both immediately useful and long-term scalable.